FINAL PROJECT SCENARIO 1

UNDERSTANDING DIABETES PROGRESSION

1.Introduction and data

- What does one row represent?

Every row represents an observation of our data. We have 442 observations, so we have 442 patients measures.

- Which variable is the outcome?

The outcome variable is Y which we renamed it as progression.

- Which variables are predictors?

All variables expect progression can be used as predictors. The predictors can be the variables: Age, sex, BMI, blood pressure and six serum measurements (s1-s6).

- Which variables are numerical, and which are categorical in meaning?

The only categorical variable is sex with two categories 1,2 representing men and women respectively. The other 10 variables are numerical.

- Are there missing values, duplicated rows or obviously problematic values?

After investigation (row 93 and 96 in the code file) the are no missing values, duplicated rows or obviously problematic values.

- Why do health-related data require careful and responsible interpretation?

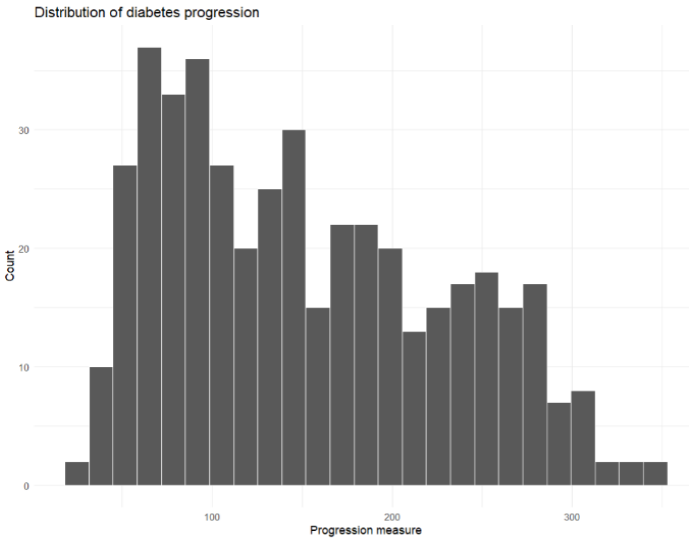
Health-related data require privacy to protect sensitive personal information, responsible use to prevent misuse or harm, and cautious interpretation to avoid misleading conclusions that could affect people's health and well-being.

2. Explore the data

- What do the centre and spread of the main variables look like?

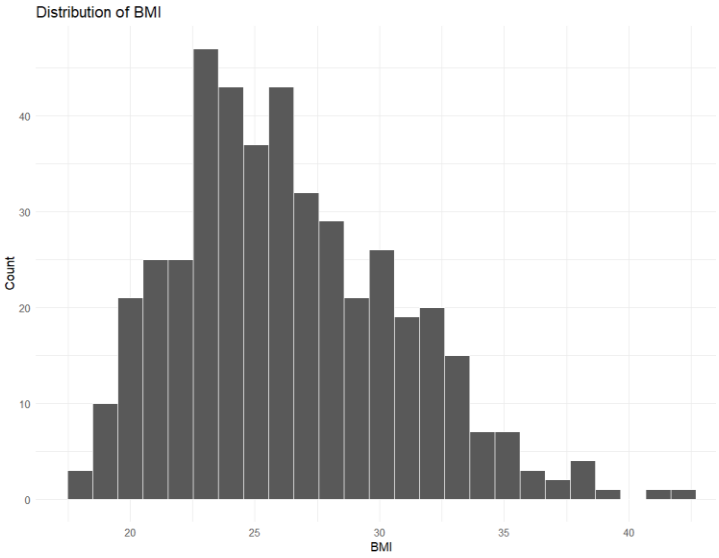
The main variables show different centres and spreads. Age has a mean of 48.52 years, median 50 years and a standard deviation of 13.11, while BMI has a mean of 26.38, median 25.7 and a standard deviation of 4.42. Diabetes progression has a mean of 152.1, a median of 140.5 and a standard deviation of 77.09. (information from descriptive_table)

- Are the distributions approximately symmetric or clearly skewed?



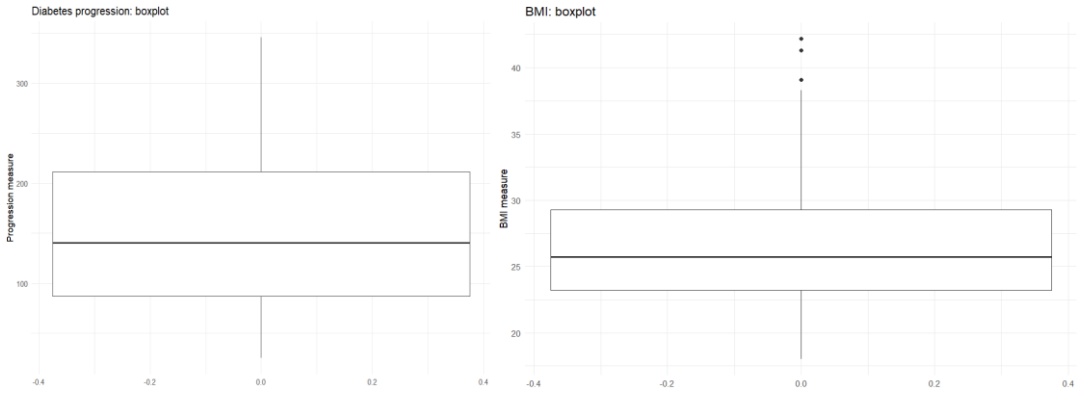
The distribution of diabetes progression appears slightly left-skewed. We can also conclude to that since the mean (152.1) is higher than the median (140.5).

The BMI distribution is approximately symmetric, although it shows a slight right skew. The mean (26.38) is close to the median (25.7), but the histogram has a longer tail towards higher BMI values.



- Are there unusual observations?

The boxplot of diabetes progression does not show any unusual observations, while the BMI boxplot identifies three unusually high values.



- Which features of the data matter before modelling begins?

Before modelling, it is important to consider the distributions of the variables, the presence of potential outliers, and the different measurement scales of the predictors. These factors may influence the choice of statistical model and whether data preprocessing, such as scaling, is necessary.

3. Investigate relationships

- Which predictors are most strongly associated with progression?

The predictors most strongly associated with diabetes progression are BMI (r=0.5865) and S5 (r=0.5659), followed by blood pressure (r=0.4415) and S4 (r=0.4305). S3 has a moderate negative correlation (r=−0.3948), while the remaining predictors show weaker positive associations.

- Which predictors are strongly related to each other?

The strongest relationships between predictors are between S1 and S2 (r=0.90), S3 and S4 (r=−0.74), and S2 and S4 (r=0.66). S4 and S5 are also positively correlated (r=0.62).

- Could some predictors carry overlapping information?

Some predictors may carry overlapping information because they are strongly correlated with each other. For example, S1 and S2 (r=0.90) and S3 and S4 (r=−0.74) show strong relationships, suggesting potential multicollinearity in regression models.

- Which variables look most promising for regression analysis, and why?

BMI and S5 look the most promising predictors for regression analysis because they have the strongest positive correlations with diabetes progression (r=0.59 and r=0.57, respectively). Blood pressure and S4 may also be useful, while S3 shows a moderate negative association (r=−0.39). Also, the strong correlations between some predictors should be considered.

4. Build and interpret regression models

- Simple linear regression

- Write the fitted equation.

The estimated coefficients of the model are intercept: -117.77 and BMI: 10.23. The equation of the fitted model is diabetes progression=-117.77+10.23*BMI.

- Interpret the slope in context.

The slope is equal to 10.23, meaning that for each one unit increase in BMI, the predicted diabetes progression increases by 10.23 units, on average.

- Does the intercept have a useful real-world interpretation?

The intercept is -117.77, representing the predicted diabetes progression when BMI is zero. However, this does not have a meaningful real-world interpretation, since a BMI of zero is unrealistic.

- What does R-squared mean here?

The R-squared value is 0.344, meaning that 34.4% of the variance in diabetes progression is explained by the linear regression model using BMI as the predictor.

- Multiple regression

- Interpret at least two coefficients from the multiple model.

- Use the phrase 'holding the other predictors constant' correctly.

The estimated coefficients of the model are intercept: -334.881, BMI: 6.5, bp: 0.903 and s5: 49.577. The equation of the fitted model is

diabetes progression= -334.881 +6.5*BMI +0.903*bp +49.577*s5.

The estimated coefficient of BMI is equal to 6.5, meaning that for each one unit increase in BMI, while holding the other predictors constant, the predicted diabetes progression increases by 6.5 units, on average. Similarly, the estimated coefficient of blood pressure is equal to 0.903, meaning that for each one unit increase in blood pressure, while holding the other predictors constant, the predicted diabetes progression increases by 0.903 units, on average.

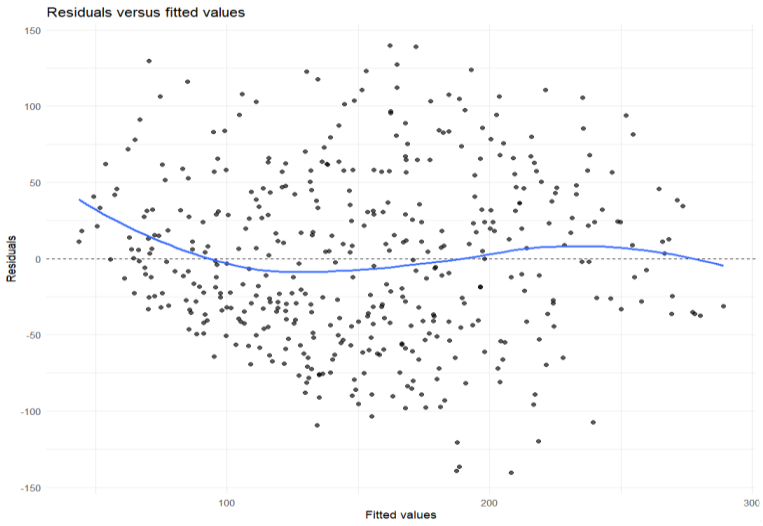
- Compare the simple and multiple models.

- Do not assume that the model with the larger R-squared is automatically better in every respect.

In order to compare the models we will use the results of the model_comparison matrix. The simple model which is using only BMI as a predictor has R-squared equal to 0.344, Adjusted R-squared equal to 0.342 and Sigma equal to 62.5. On the other hand, the multiple model which is using BMI, blood pressure and s5 as a predictors has an R-squared of 0.48, an Adjusted R-squared of 0.477 and a Sigma of 55.8. The multiple model performs better in terms of explained variation, as its R-squared increases from 0.344 to 0.480. This means that the multiple model explains 48.0% of the variance in diabetes progression, compared with 34.4% for the simple one. The adjusted R-squared also increases from 0.342 to 0.477, indicating that the improvement remains after accounting for the number of predictors. Furthermore, sigma decreases from 62.5 to 55.8, suggesting that the prediction errors are smaller in the multiple model. Overall, the multiple model appears to provide a better fit to the data. However, a higher R-squared alone does not guarantee that a model is better in every respect, since model complexity and predictive performance should also be considered.

5. Evaluate the analysis

- Comment on the residuals-versus-fitted plot.



The residuals are generally scattered around zero, forming a cloud of points without a strong systematic pattern. However, the smoothed blue line shows a slight curve, which may indicate some non-linearity in the relationship. Overall, the plot does not reveal a strong violation of the linearity assumption, although the model may not capture all patterns in the data.

- Use VIF and the correlation matrix to discuss multicollinearity.

The VIF values are 1.345 for BMI, 1.274 for blood pressure, and 1.342 for s5. These values are close to 1, indicating that there is no serious multicollinearity among the predictors in the multiple regression model. Although the correlation matrix showed strong correlations between some of the original predictors, the three predictors selected for this model do not appear to have a multicollinearity problem.

- For at least two important coefficients, discuss the estimate and either the confidence interval, t-value or p-value.

BMI, blood pressure, and s5 are all statistically significant predictors of diabetes progression, as their p-values are below 0.001. The estimated coefficient of BMI is 6.50, with a 95% confidence interval from 5.13 to 7.87. Blood pressure has an estimated coefficient of 0.903, with a 95% confidence interval from 0.477 to 1.33. S5 also has a positive estimated coefficient of 49.6, with a 95% confidence interval from 38.0 to 61.2. Since all three confidence intervals exclude zero, the results support positive associations between each predictor and diabetes progression, holding the other predictors constant.

- Distinguish statistical significance from practical importance.

Statistical significance indicates that the observed associations are unlikely to be due to random sampling variation alone, under the model assumptions. However, it does not necessarily mean that the effects are practically important. The practical importance of each predictor depends on its measurement scale and on whether the size of its estimated effect is meaningful in the context of diabetes progression. Therefore, the coefficients should be interpreted in terms of their units and real-world relevance, rather than relying only on their p-values.

- State the main statistical weaknesses or uncertainties in the analysis.

One limitation of the analysis is that the model explains only 48.0% of the variance in diabetes progression, so a substantial proportion remains unexplained. The residual plot

shows considerable variability and a slight curved pattern, suggesting that the linear model may not capture every aspect of the relationship. In addition, the analysis identifies associations rather than proving causation. The conclusions also depend on the assumptions of the regression model.

6. Draw appropriate conclusions

The dataset is observational rather than experimental. Therefore, the regression coefficients show associations between the predictors and diabetes progression, but they cannot be interpreted as proof of causal relationships. There may be confounding variables that are not included in the model, such as diet or medication. These variables could influence diabetes progression and may affect the estimated relationships. Measurement errors or limitations in the way variables were recorded could also influence the results. The findings may not necessarily generalize to other populations, since the results depend on the characteristics of the individuals included in the dataset. The evidence supports the conclusion that BMI, blood pressure, and S5 are statistically significant predictors of diabetes progression in this dataset, with positive estimated associations when the other predictors are held constant. The multiple regression model explains 48.0% of the variance in diabetes progression. However, claiming that increasing BMI or blood pressure directly causes diabetes progression, that the model explains all relevant factors, or that the same relationships apply to every population would go beyond the available evidence.